Transient Photoresponse Analysis of Quantum Dot Photodetectors for Ultra Fast Imaging

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***Abstract*—Transient photoresponse characteristics of quantum dot (QD) photodetectors are critical for their application in ultra-fast imaging systems. In this work, we systematically analyze the response dynamics of PbS QD photodiodes under varying optical power levels. Our measurements demonstrate a clear dependence of transient response speed on light intensity, where higher optical powers lead to slower response, manifested by prolonged rise times. This phenomenon aligns with bimolecular recombination dynamics and is attributed to increased photogenerated carrier densities that enhance recombination rates and induce space-charge effects, impeding charge separation and transport. These insights deepen the understanding of transient behaviors in QD-based photodetectors and provide guidance for mitigating recombination effects to advance high-speed optoelectronic imaging applications.**

Keywords—quantum Dot, PbS, photodiodes, response speed, light intensities

I. Introduction

Photodiodes with high-speed response are integral to a wide range of cutting-edge photonic technologies. These devices leverage the inherent advantages of optical systems, such as high bandwidth, low noise, and minimal signal loss, ensuring excellent optical performance. High-speed photodiodes made from PbS quantum dot materials further enhance the response speed and operational efficiency across various applications. They have already been widely adopted in microwave-photonic links, opto-electronic oscillators (OEOs), photon-based radar, photonic signal processing, high-speed waveform generation, and the generation of low-noise microwaves via optical frequency division (OFD) [1].

In this study, we investigate the response speed of photodiodes fabricated from lead sulfide (PbS) quantum dots, a material known for its promising optical and electronic properties. PbS quantum dots have attracted significant attention due to their tunable bandgap, high absorption efficiency, and potential for integration into high-performance optoelectronic devices.

Our experiments reveal a inverse proportionality between the response speed of PbS quantum dot photodiodes and varying light intensities. This paper aims to provide a detailed explanation of the underlying mechanisms responsible for this relationship, shedding light on the interaction between light intensity and the photodiode response, as well as the role of the quantum dot material in influencing these dynamics.

II. Experiment Results

The quantum dot photodetector used in the experimental study is designed with a layered structure, which includes, from top to bottom: Ag (silver) electrode, MoOx, PbS-EDT (PbS quantum dot surface passivation), PbS-ink (PbS quantum dot film), ZnO, ITO, Glass (glass substrate) and other layers as shown in Figure 1.

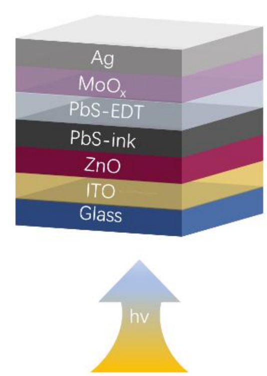

Fig. 1. Photodiode Design Structure.

Firstly, we characterized the performance of the photodiode produced as shown in Figure 2 and Figure 3.

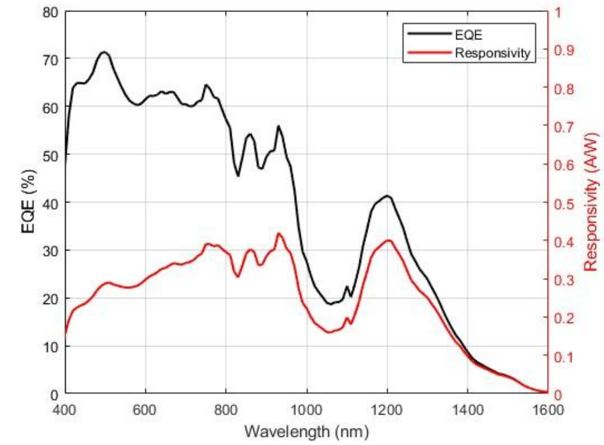

Fig. 2. EQE and Responsivity Characteristics.

Figure 2 shows the spectral response characteristics of the photodiode in the 400–1600 nm range. This figure includes curves for external quantum efficiency and responsivity. Both the EQE and responsivity curves remain relatively high near 1310 nm. 1310 nm, as a classical optical frequency, demonstrates the device's broadband absorption capability, making it suitable for SWIR imaging applications.

Figure. 3 illustrates the JV characteristics of the photodiode under two illumination conditions: 4.8 mW (black curve) and 48 μW (red curve). The current density is significantly higher at 4.8 mW across the entire voltage range, indicating that higher optical power generates a greater number of electron-hole pairs, thus increasing the photocurrent. Additionally, the open-circuit voltage (Voc) is slightly higher under higher illumination, consistent with the increased carrier density and enhanced quasi-Fermi level splitting. In the reverse bias region, the dark current remains stable for the 48 μW case, while the 4.8 mW curve exhibits a slight deviation, possibly due to increased leakage currents or stronger carrier injection under higher power.

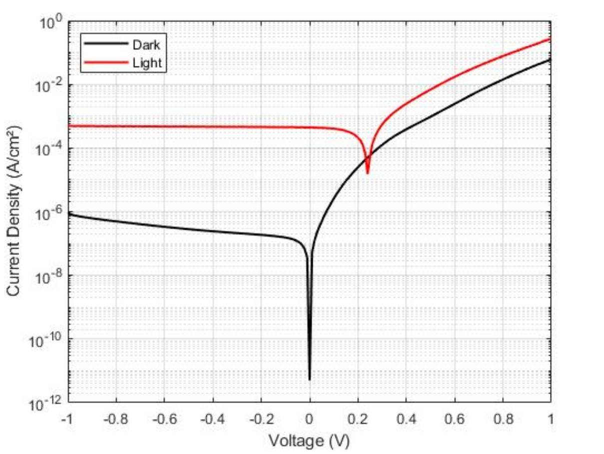

Fig. 3. JV Characteristics.

Furthermore, the asymmetry in the JV curves near 0 V becomes more pronounced under dark (4.8 mW) illumination, reflecting shifts in the photodiode's operating point induced by higher light intensity. These results highlight the photodiode's strong dependence on optical power for its electrical characteristics, with higher power amplifying photocurrent and Voc. However, the observed non-linearities and deviations under high illumination suggest potential impacts of material or structural limitations, which merit further investigation to optimize device performance under varying illumination conditions.

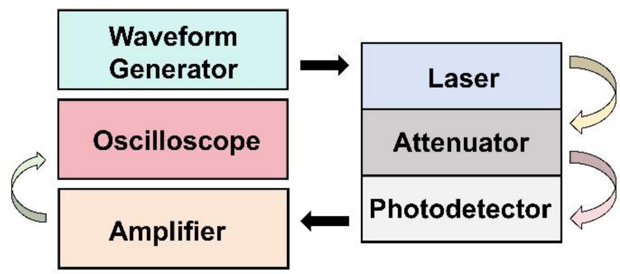

Fig. 4. Optical Test System.

In the experiment, an optical setup was constructed with a stable 1310 nm wavelength laser source providing the excitation, a waveform generator, a tunable attenuator, a low-noise amplifier and an oscilloscope, as shown in Figure 4. Optical attenuation was achieved using an NDFA10-20-C absorptive neutral density filter (OD-2.0, with a 1100 nm to 1650 nm transmission range and a 25.4 mm diameter), which allowed for precise control of the light power. While tunable attenuator as the backup device. An optical amplifier was used to enhance the signal after the photoelectric conversion. The amplified signal was monitored using a high-precision optical oscilloscope. To facilitate power measurements, a power meter was used to measure the light intensity. The maximum power measured was 4.8 mW, and for analysis, we focused on the operating conditions at two power levels: 4.8mW and 48uW, as shown in Figure 5.

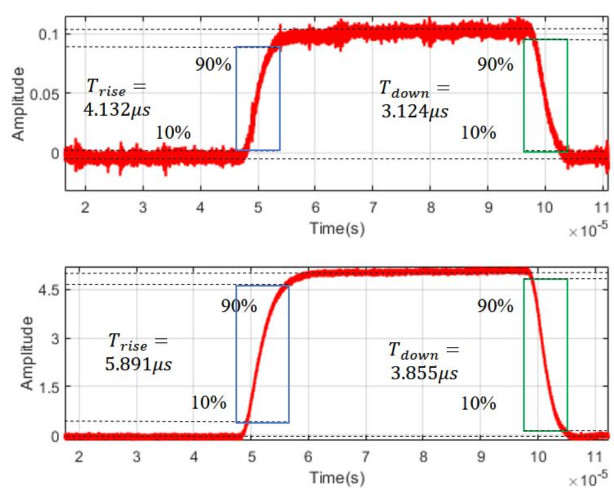

Fig. 5. Transient response at different optical powers.

The upper curve, shown in blue, represents the relationship between current amplitude and time in the optical amplifier at a light power of 4.8 mW, while the lower curve, shown in red, corresponds to a power of 48 μW after attenuation by the optical attenuator.

To assess the response speed, we measured the time taken for the first rising edge after 0 μs, specifically focusing on the rise time. For the lower power case (48 μW), the measured rise time is 4.132 μs, whereas for the higher power case (4.8 mW), the rise time is 5.891 μs. These results indicate that, despite the increased light intensity, the rise time at higher power is slightly longer, which suggests that response speed decreases with increasing optical power under the conditions tested.

III. THEORETICAL UNDERSTANDING

It can be seen from the experimental results that when the light power increased, the response speed of the photodiode decreased accordingly, which was exactly the same as the results obtained by previous experiments [2,3]. The difference between this experiment and previous experiments is that they increased the optical power by adjusting the reverse bias voltage, while we increased the optical power by adding an optical attenuator to control the light intensity. In the literature, many studies [4] adjust the response speed of optical devices by varying the reverse bias voltage to change the optical power. However, in our experiment, we manipulated the optical power by introducing the optical attenuator. Although both methods aim to influence the optical power and, consequently, the response speed, they operate through different physical mechanisms. The reverse bias voltage directly influences the electrical characteristics of the device, such as the drift current in photodiodes, thereby affecting the response time. In contrast, the optical attenuator reduces the intensity of the incident light, indirectly affecting the interaction between the light and the material, which in turn influences properties like the intensity and propagation mode of the light, and consequently, the response speed. Therefore, these two approaches are fundamentally different in how they impact the device's performance.

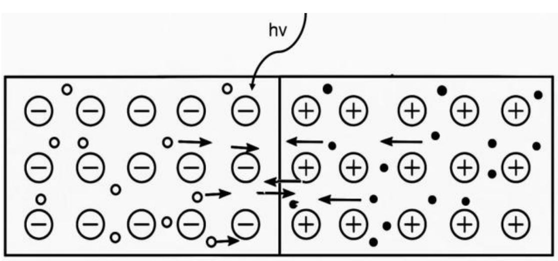

Fig. 6. Bimolecular recombination. (*figure caption*)

The impact of optical power on the response speed of optoelectronic devices can be explained through the process of bimolecular recombination, a fundamental mechanism in semiconductors. When optical power is increased, more photons are absorbed by the material, generating a larger number of electron-hole pairs. These charge carriers contribute to the device's current, but they do not remain in this excited state indefinitely. Eventually, the electron-hole pairs recombine, releasing energy in the form of heat or light. The rate at which this recombination occurs, known as the recombination rate, plays a critical role in the overall performance of the device, particularly its response speed [5].

Bimolecular recombination refers to the process in which an electron and a hole meet and recombine, effectively neutralizing each other. This recombination rate is proportional to the product of the electron and hole densities, typically expressed as:

$$R = \beta np \tag{1}$$

where R is the recombination rate, β is the recombination coefficient, n is the electron concentration, and p is the hole concentration.

$$t_{rise/fall} \propto \lambda = \frac{1}{Bn} = \sqrt{\frac{hv}{\eta BI}} \tag{2}$$

Here, t_(rise/fall) is the response time, λ is the carrier lifetime, B is the bimolecular recombination coefficient, n is the carrier concentration, hv is the photon energy, η is the quantum efficiency, and I is the optical intensity.

The above formulas are derived from the literature [5-6]. When optical power increases, the photon absorption leads to an increase in the electron-hole pair generation, thereby increasing the carrier concentrations n and p. This, in turn, leads to a higher recombination rate. As the recombination rate increases, the carrier lifetime (λ) decreases. This shorter carrier lifetime means that the electrons and holes spend less time before recombining, limiting the amount of time available for charge carriers to contribute to the current. As a result, the overall response speed of the device is slowed down because the current can no longer accumulate as quickly. This process can be explained by Figure 6.

Under typical conditions, increasing optical power would result in higher carrier densities, which leads to an increased recombination rate and a reduced carrier lifetime. As a result, the response speed of the device would decrease because the carriers have less time to accumulate and contribute to the current. This is the standard expectation based on bimolecular recombination theory.

The observed strong dependence of the device responsivity on incident optical power is consistent with the behavior typically seen in ligand-exchanged PbS CQD photoconductors [7]. At low optical power levels, high photoconductive gain can be attributed to efficient electron capture by mid-gap trap states, as the photogenerated carrier density remains below the trap density. However, with increasing light intensity, the gradual filling of trap states raises the electron quasi-Fermi level, reducing the capture rate and subsequently lowering the gain. This mechanism highlights the significant influence of trap dynamics on the performance of CQD-based devices. Such findings are in agreement with previously reported studies, which have demonstrated similar power-dependent responsivity trends and attributed the high responsivity to the balance between electron lifetime and transit time across the device.

This conclusion aligns with our experimental results, where we observed that as optical power increased, the response speed decreased. This finding is consistent with the predictions of the bimolecular recombination model, which suggests that higher optical power leads to increased carrier density, resulting in prolonged carrier lifetimes and slower recombination dynamics. Unlike light-emitting devices where recombination is desirable for photon generation, our device operates on the opposite principle: we aim to minimize recombination and instead maximize the time carriers remain in the excited state, allowing them to be efficiently extracted by the built-in electric field or an applied bias.

In this context, the slower response speed at higher optical powers can be attributed to the higher carrier density leading to space charge effects, which hinder the extraction of carriers. The increased carrier concentration reduces the efficiency of charge separation and transport, as carriers are more likely to screen the internal electric field. This underscores the critical role of carrier dynamics in non-light-emitting devices, where the goal is to suppress recombination and optimize the extraction process. Therefore, the observed results not only align with the bimolecular recombination model but also highlight the importance of device design in managing carrier transport under varying optical power conditions.

IV. CONCLUSION

This study systematically investigated the transient photoresponse of PbS quantum dot photodiodes under varying optical power levels. We observed that higher illumination intensities led to slower response speeds, characterized by prolonged rise times. This behavior is consistent with bimolecular recombination dynamics, where increased carrier densities enhance recombination rates and induce space-charge effects, thereby impeding charge separation and transport. Additionally, trap-state dynamics intrinsic to PbS CQD photoconductors were evident, with low-power excitation favoring higher photoconductive gain through efficient carrier trapping, while high-power excitation led to trap saturation. These findings elucidate the interplay between recombination processes, trap dynamics, and optical power in governing device speed. They also provide valuable design insights for optimizing QD-based photodetectors, particularly for ultra-fast imaging applications where maintaining high-speed performance under varying illumination is critical. Future work will focus on material engineering and structural innovations to suppress recombination losses and mitigate trap saturation, enabling broader applicability in high-speed photonic systems.

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